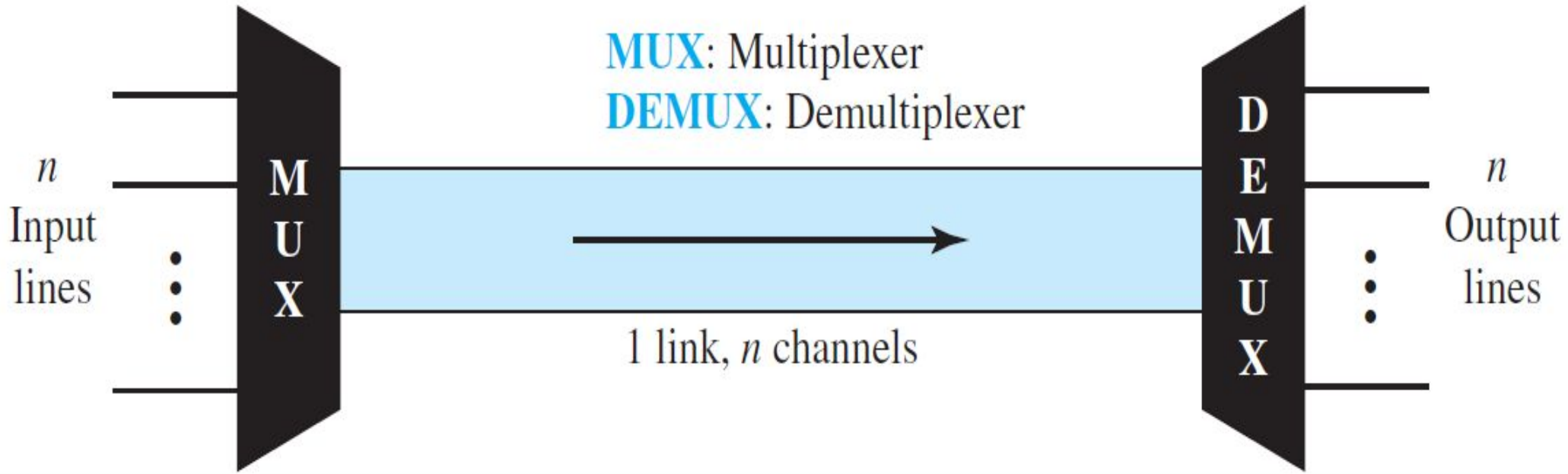
# Bandwidth Utilization: Multiplexing and Spectrum Spreading

## MULTIPLEXING

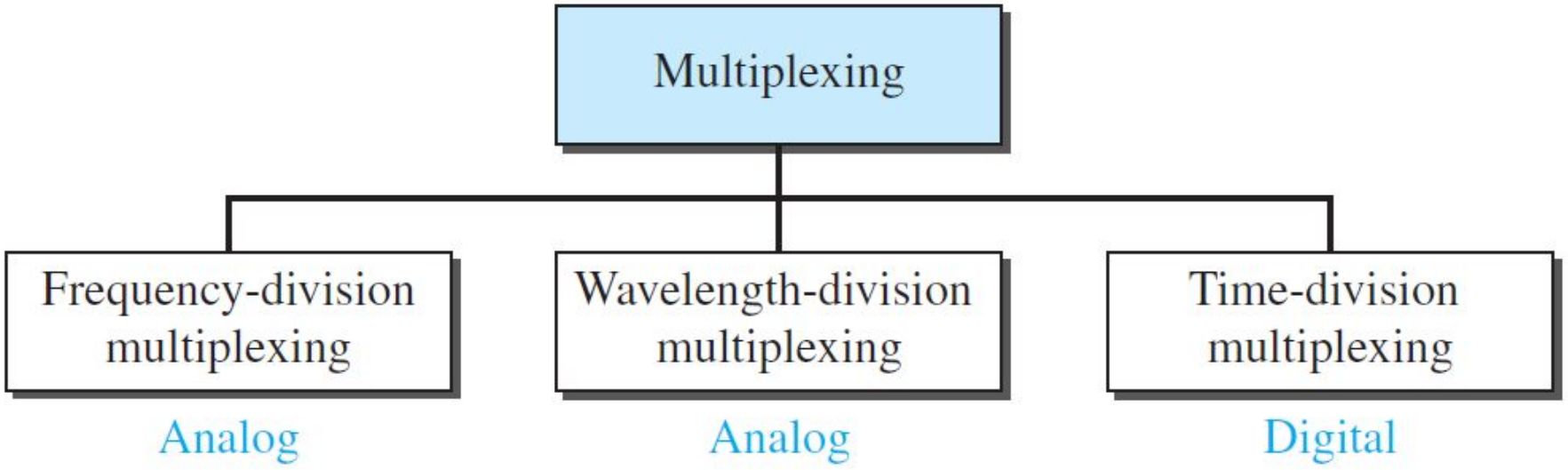
- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth
- needs of the devices, the link can be shared.
- **Multiplexing** is the set of technique that allow the simultaneous transmission of multiple signals across a single data link.
- As data and telecommunications use increases, so does traffic.
- Accommodate this increase by continuing to add individual links each time a new channel is needed; or we can install higher-bandwidth links and use each to carry multiple signals
- In a multiplexed system, *n* lines share the bandwidth of one link.
- The lines on the left direct their transmission streams to a **multiplexer (MUX),** which combines them into a single stream (many-toone).
- At the receiving end, that stream is fed into a **demultiplexer (DEMUX),** which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.
- the word **link** refers to the physical path.
- The word **channel** refers to the portion of a link that carries a transmission between a given pair of lines. One link can have many (*n*) channels.

**Figure 6.1** *Dividing a link into channels*



There are three basic multiplexing techniques: frequency-division multiplexing, wavelength-division multiplexing, and time-division multiplexing. The first two are techniques designed for analog signals, the third, for digital signals

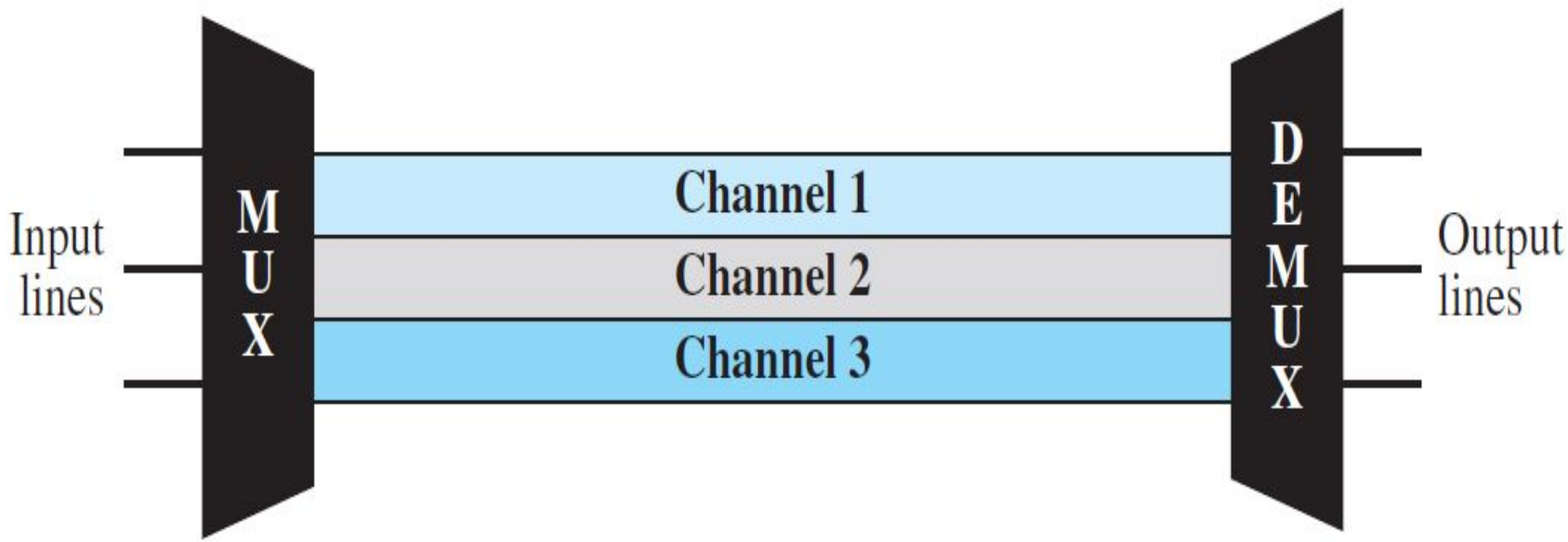
**Figure 6.2** *Categories of multiplexing*



# Frequency-Division Multiplexing

- **Frequency-division multiplexing (FDM)** is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.
- In FDM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link.
- Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal.
- These bandwidth ranges are the channels through which the various signals travel. Channels can be separated by strips of unused bandwidth—**guard bands**—to prevent signals from overlapping
- FDM to be an analog multiplexing technique; however, this does not mean that FDM cannot be used to combine sources sending digital signals.
- A digital signal can be converted to an analog signal before FDM is used to multiplex them

**Figure 6.3** *Frequency-division multiplexing*

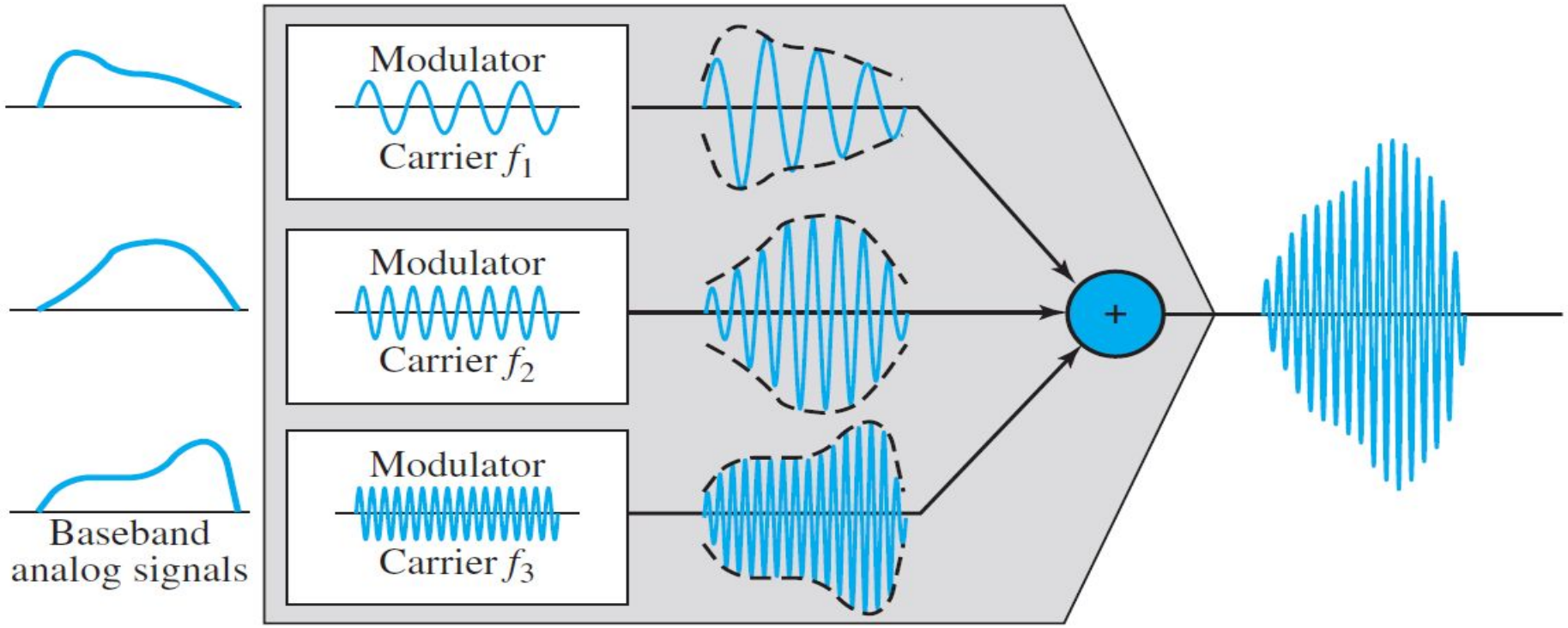


# MULTIPLEXING PROCESS

Each source generates a signal of a similar frequency range. Inside the multiplexer, these similar signals modulate different carrier frequencies ( $f1$, $f2$, and $f3$). The resulting modulated signals

are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

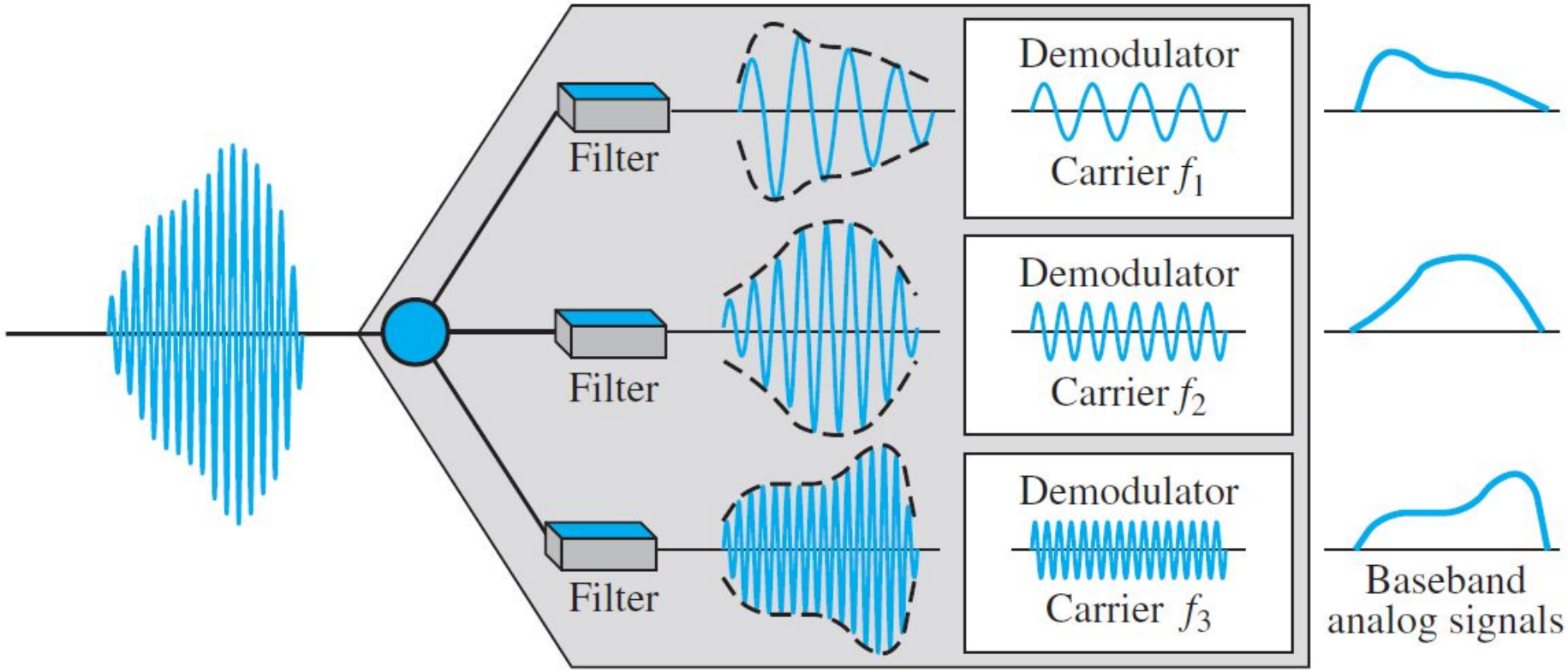
**Figure 6.4** *FDM process*



# *Demultiplexing Process*

The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals. The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines.

**Figure 6.5** *FDM demultiplexing example*



**Example 6.1**

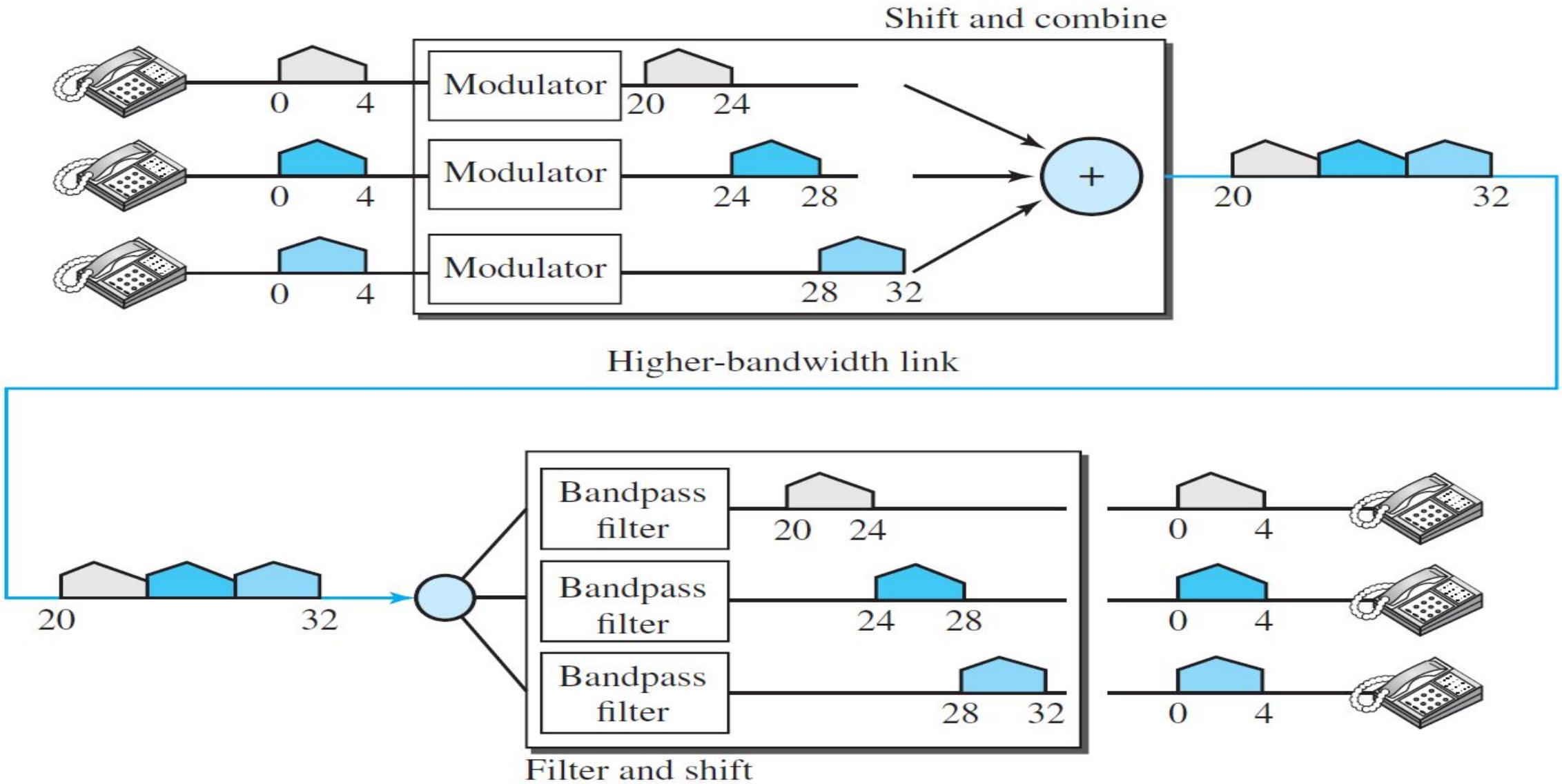
Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

**Solution**

We shift (modulate) each of the three voice channels to a different bandwidth, We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them At the receiver, each channel receives the entire signal, using a filter to separate out its own signal.

The first channel uses a filter that passes frequencies between 20 and 24 kHz and filters out (discards) any other frequencies.

The second channel uses a filter that passes frequencies between 24 and 28 kHz, and the third channel uses a filter that passes frequencies between 28 and 32 kHz. Each channel then shifts the frequency to start from zero.

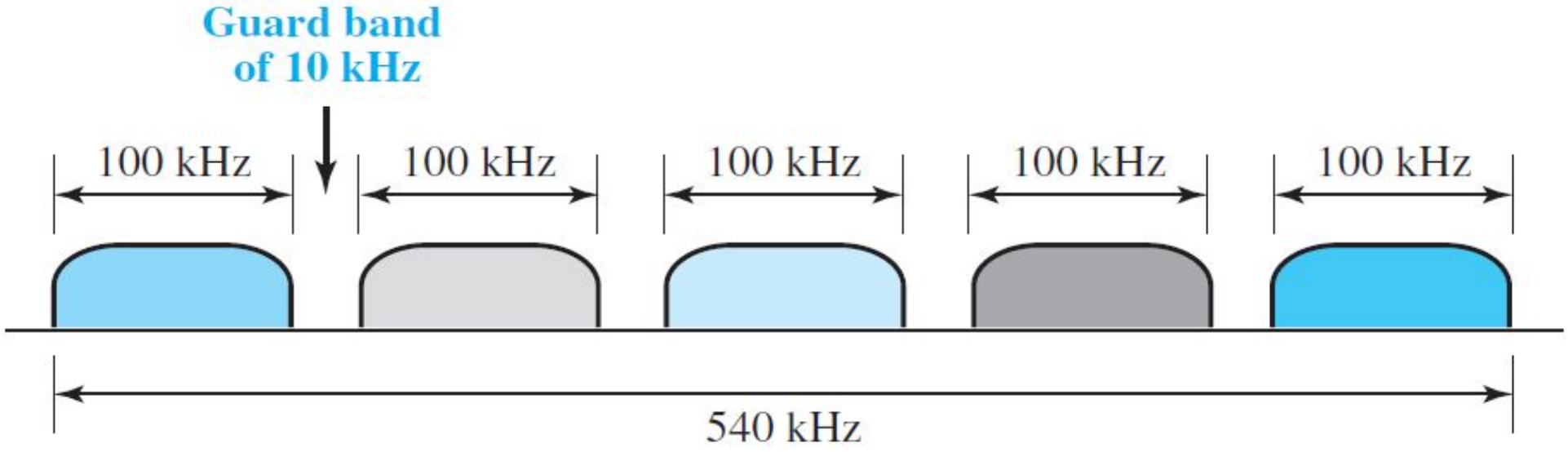

Shift and combine
Modulator
Modulator
Modulator
0
4
20
24
28
32
+
Higher-bandwidth link
Bandpass filter
Bandpass filter
Bandpass filter
Filter and shift

Example 6.2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is atleast 5 × 100 + 4 × 10 = 540 kHz,

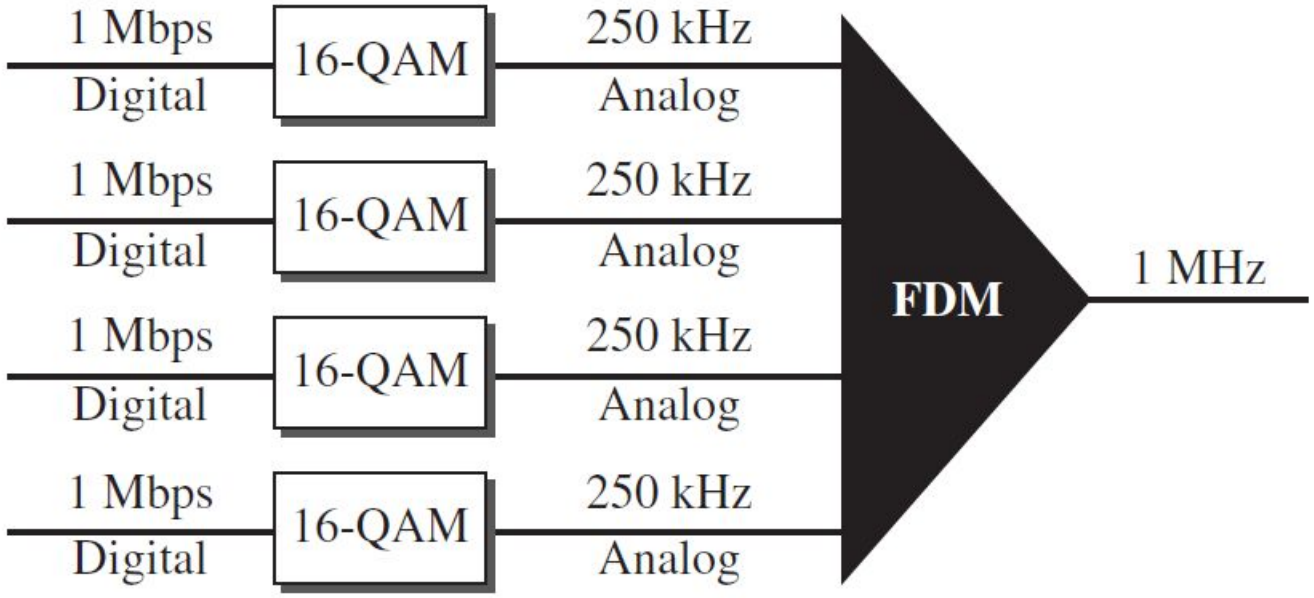


Example 6.3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.

Solution

The satellite channel is analog. We divide it into four channels, each channel having a 250-kHz bandwidth. Each digital channel of 1 Mbps is modulated so that each 4 bits is modulated to 1 Hz. One solution is 16-QAM modulation.



## *The Analog Carrier System*

To maximize the efficiency of their infrastructure, telephone companies have traditionally

multiplexed signals from lower-bandwidth lines onto higher-bandwidth lines. In

this way, many switched or leased lines can be combined into fewer but bigger channels.

For analog lines, FDM is used.

One of these hierarchical systems used by telephone companies is made up of groups, supergroups, master groups, and jumbo groups

**analog hierarchy,** 12 voice channels are multiplexed onto a higher-bandwidthline to create a **group.** A group has 48 kHz of bandwidth and supports 12 voice channels.

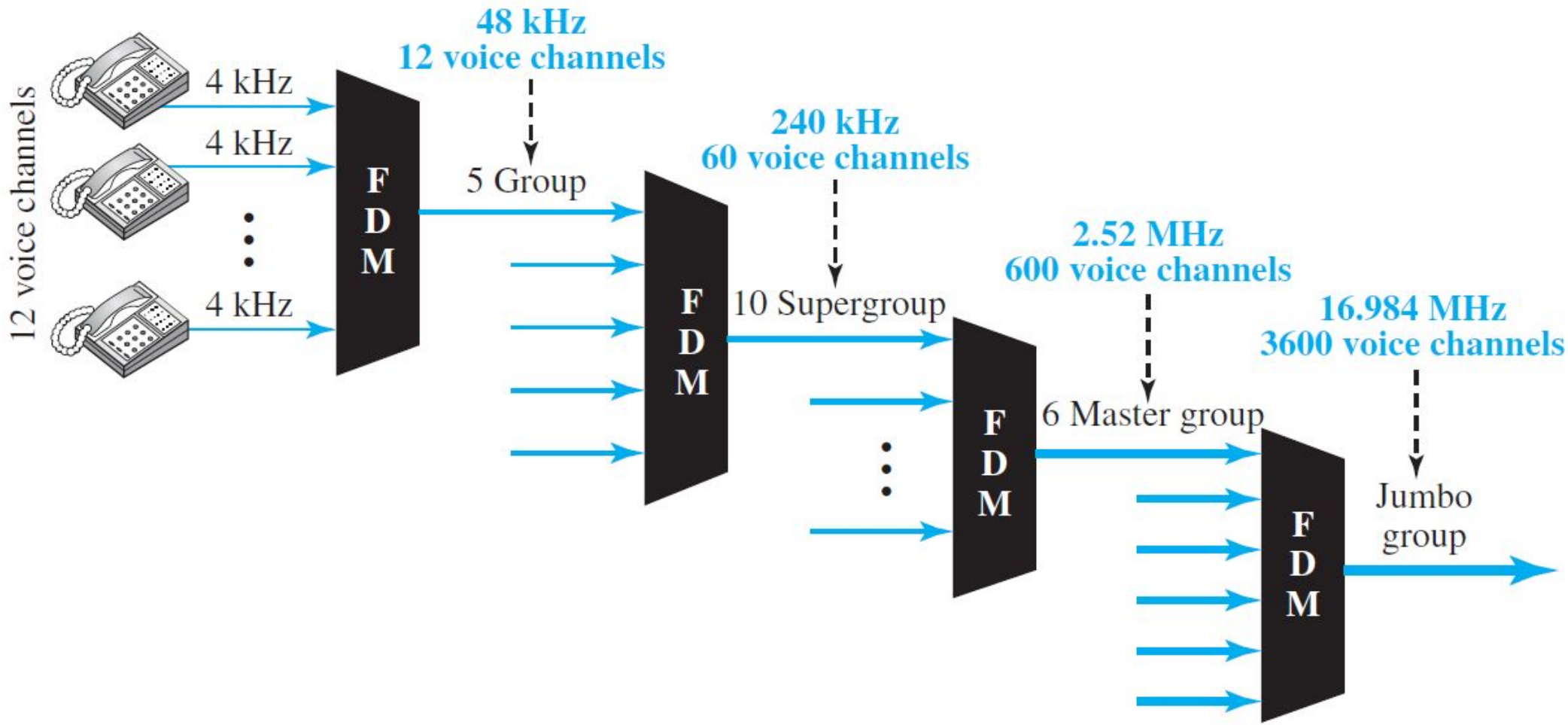
## *The Analog Carrier System*

- At the next level, up to five groups can be multiplexed to create a composite signal called a **supergroup.** A supergroup has a bandwidth of 240 kHz and supports up to 60 voice channels. Supergroups can be made up of either five groups or 60 independent voice channels.

- At the next level, 10 supergroups are multiplexed to create a **master group.** A master group must have 2.40 MHz of bandwidth, but the need for guard bands between the supergroups increases the necessary bandwidth to 2.52 MHz. Master groups support up to 600 voice channels.

- six master groups can be combined into a **jumbo group.** A jumbo group must have 15.12 MHz (6 × 2.52 MHz) but is augmented to 16.984 MHz to allow forguard bands between the master groups.

**Figure 6.9** *Analog hierarchy*



# Applications of FDM

- A very common application of FDM is AM and FM radio broadcasting. Radio uses the air as the transmission medium. A special band from 530 to 1700 kHz is assigned to AM radio. All radio stations need to share this band. Each AM station needs 10 kHz of bandwidth.
- Each station uses a different carrier frequency, which means it is shifting its signal and multiplexing.
- The signal that goes to the air is a combination of signals. A receiver receives all these signals, but filters (by tuning) only
- the one which is desired.
- Without multiplexing, only one AM station could broadcast to the common link, the air. However, we need to know that there is no physical multiplexer or demultiplexer here. multiplexing is done at the data-link layer.
- The situation is similar in FM broadcasting. .

Applications of FDM

- FM has a wider band of 88 to 108 MHz because each station needs a bandwidth of 200 kHz.
- Another common use of FDM is in television broadcasting. Each TV channel has its own bandwidth of 6 MHz.
- The first generation of cellular telephones  also uses FDM.
- Each user is assigned two 30-kHz channels, one for sending voice and the other for receiving.
- The voice signal, which has a bandwidth of 3 kHz (from 300 to 3300 Hz), is modulated by using FM.
- that an FM signal has a bandwidth 10 times that of the modulating signal, which means each channel has 30 kHz (10 × 3) of bandwidth.
- Therefore, each user is given, by the base station, a 60-kHz bandwidth in a range available at the time of the call.

- The Advanced Mobile Phone System (AMPS) uses two bands. The first band of 824 to 849 MHzis used for sending, and 869 to 894 MHz is used for receiving. Each user has a bandwidth of30 kHz in each direction. The 3-kHz voice is modulated using FM, creating 30 kHz of modulated signal. How many people can use their cellular phones simultaneously?
- **Solution**
- Each band is 25 MHz. If we divide 25 MHz by 30 kHz, we get 833.33. In reality, the band is divided into 832 channels. Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users.

- ***Implementation***
- FDM can be implemented very easily. In many cases, such as radio and television

- broadcasting, there is no need for a physical multiplexer or demultiplexer. As long as the stations agree to send their broadcasts to the air using different carrier frequencies, multiplexing is achieved.

- In other cases, such as the cellular telephone system, a base station needs to assign a carrier frequency to the telephone user. There is not enough bandwidth in a cell to permanently assign a bandwidth range to every telephone user.

- When a user hangs up, her or his bandwidth is assigned to another caller.

- **Wavelength-Division Multiplexing**
- **Wavelength-division multiplexing (WDM)** is designed to use the high-data-rate capability of fiber-optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable, but using a fiber-optic cable for a single line wastes the available bandwidth. Multiplexing allows us to combine several lines into one.
- WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- The idea is the same: We are combining different signals of different frequencies. The difference is that the frequencies are very high.
- Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer
- The combining and splitting of light sources are easily handled by a prism.

Figure 6.10 *Wavelength-division multiplexing*

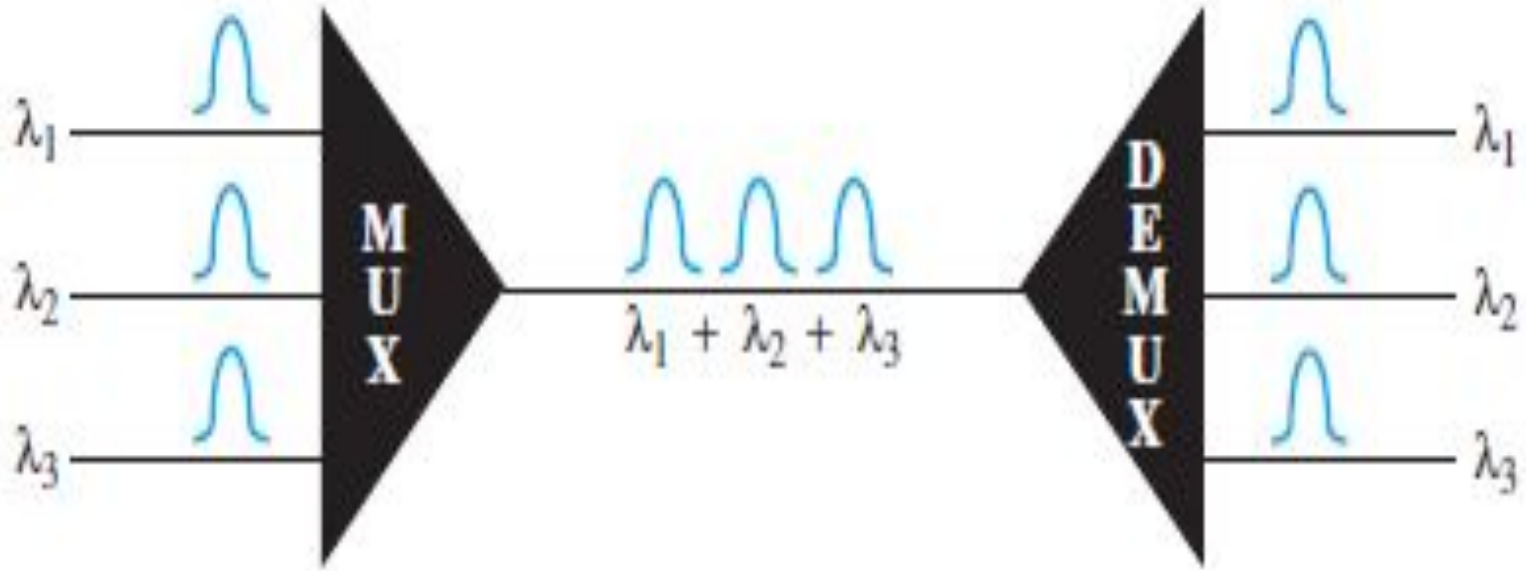
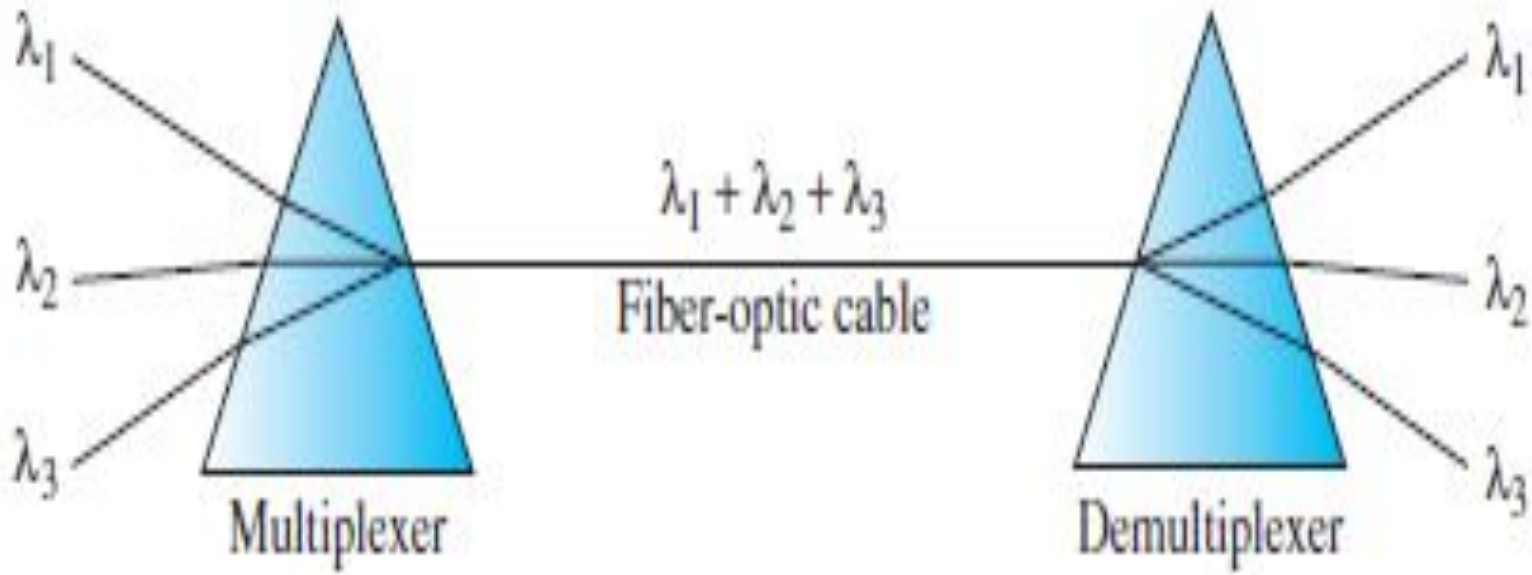


**Figure 6.11** *Prisms in wavelength-division multiplexing and demultiplexing*



# Time-Division Multiplexing

- **Time-division multiplexing (TDM)** is a digital process that allows several connections
- to share the high bandwidth of a link. Instead of sharing a portion of the bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link.

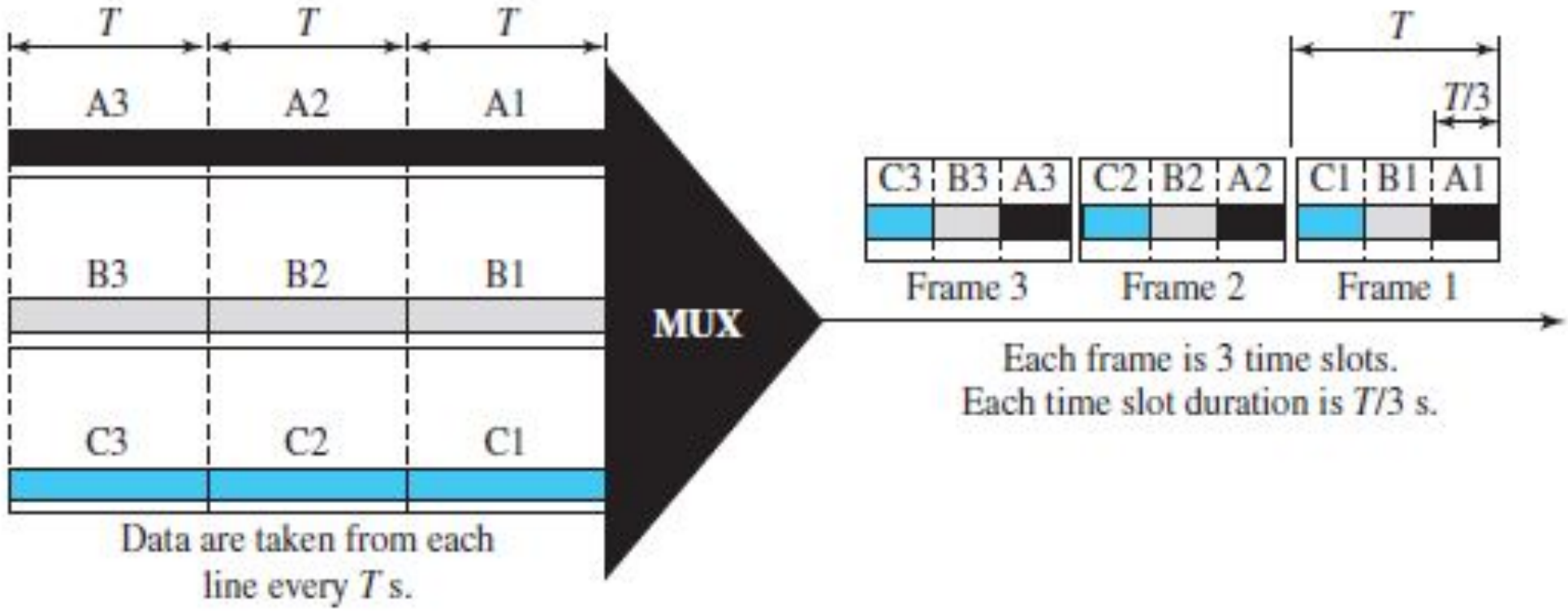
Figure 6.12 *TDM*



- concerned with only multiplexing, not switching. This means that all the data in a message from source 1 always go to one specific destination, be it 1, 2, 3, or 4.
- The delivery is fixed and unvarying, unlike switching.
- TDM is, in principle, a digital multiplexing technique.
- Digital data from different sources are combined into one timeshared link
- does not mean that the sources cannot produce analog data; analog data can be sampled, changed to digital data, and then multiplexed by using TDM.
- TDM into two different schemes: synchronous and statistical.

- ***Synchronous TDM***
- In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.
- ***Time Slots and Frames***
- In synchronous TDM, the data flow of each input connection is divided into units,
- where each input occupies one input time slot. A unit can be 1 bit, one character, or one block of data.
- Each input unit becomes one output unit and occupies one output time slot.
- However, the duration of an output time slot is *n* times shorter than the duration of an input time slot.
- If an input time slot is *T* s, the output time slot is *T*/*n* s, where *n* is the number of connections.
- a unit in the output connection has a shorter duration; it travels faster

**Figure 6.13** *Synchronous time-division multiplexing*



- In Time Division Multiplexing (TDM), multiple input signals share a single communication channel by transmitting data in different time slots.
- **1. Input Bit Duration**
- Definition: Input bit duration is the time required to transmit one bit from an individual input source to the multiplexer.
- Formula:

$$T_{\text{in}} = \frac{1}{R_{\text{in}}}$$

where $R_{\text{in}}$ is the input bit rate in bits per second (bps).

**Example:** If the input bit rate is 1 kbps (1000 bps),

$$T_{\text{in}} = \frac{1}{1000} = 1 \text{ ms}$$

## 2. Output Bit Duration

**Definition:** Output bit duration is the time required to transmit one bit on the multiplexed output channel.

**Formula:**

$$T_{\text{out}} = \frac{1}{R_{\text{out}}}$$

**Example:** If four input channels, each operating at 1 kbps, are multiplexed without overhead,

$$R_{\text{out}} = 4 \times 1000 = 4000 \text{ bps}$$

$$T_{\text{out}} = \frac{1}{4000} = 0.25 \text{ ms}$$

Therefore, output bit duration = **0.25 ms**.

### 3. Output Bit Rate

**Definition:** Output bit rate is the total number of bits transmitted per second through the multiplexed output channel.

**Formula (for equal-rate inputs without overhead):**

$$R_{\text{out}} = n \times R_{\text{in}}$$

where $n$ is the number of input channels.

**Example:** Four input channels, each transmitting at 1 kbps:

$$R_{\text{out}} = 4 \times 1 = 4\ \text{kbps}$$

**4. Frame Rate**

**Definition:** Frame rate is the number of TDM frames transmitted per second. A frame consists of time slots allocated to the input channels.

**Formula:**

$$F = \frac{R_{\text{out}}}{\text{Number of bits per frame}}$$

**Example:** If four input channels contribute one bit each per frame, the frame contains 4 bits.

$$F = \frac{4000}{4} = 1000 \text{ frames/sec}$$

Therefore, frame rate = **1000 frames per second**.

- In synchronous TDM, a round of data units from each input connection is collected into a frame .
- If we have n connections, a frame is divided into n time slots and one slot is allocated for each unit, one for each input line.
- If the duration of the input unit is T, the duration of each slot is T/n and the duration of each frame is T
- The data rate of the output link must be n times the data rate of a connection to guarantee the flow of data.
- Time slots are grouped into frames. A frame consists of one complete cycle of

time slots, with one slot dedicated to each sending device. In a system with *n* input lines, each frame has *n* slots, with each slot allocated to carrying data from a specific input line.

- In fig 6.13 the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of
- **1.** each input slot,
- **2.** each output slot, and
- **3.** each frame?
- **Solution**

**1.** The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 sor 1 ms. The duration of the input time slot is 1 ms (same as bit duration).

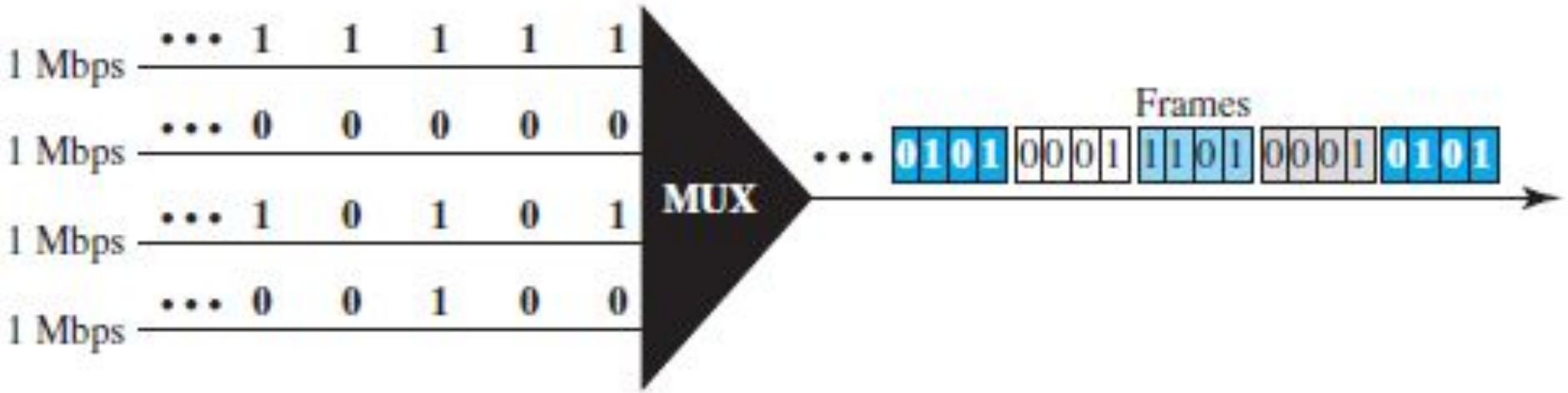
**2.** The duration of each output time slot is one-third of the input time slot. This means that theduration of the output time slot is 1/3 ms.

**3.** Each frame carries three output time slots. So the duration of a frame is 3 × 1/3 ms, or 1 ms.

The duration of a frame is the same as the duration of an input unit.

- **Example 6.6**
- Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (1) the input bit duration, (2) the output bit duration, (3) the output bit rate, and (4) the output frame rate.

Figure 6.14 *Example 6.6*

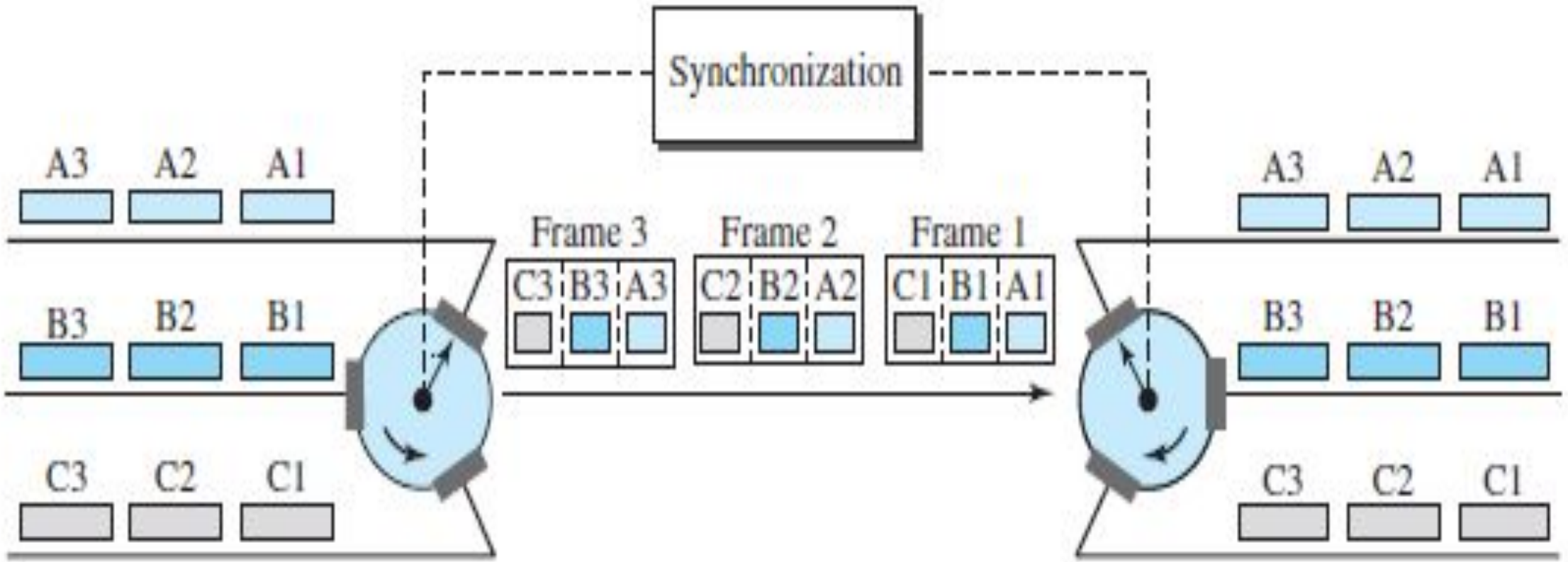


- **Solution**
- **1.** The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 µs.
- **2.** The output bit duration is one-fourth of the input bit duration, or 1/4 µs.
- **3.** The output bit rate is the inverse of the output bit duration, or 1/4 µs, or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.
- **4.** The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame,
- verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

- **Example 6.7**
- Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (1) the duration of 1 bit
- before multiplexing, (2) the transmission rate of the link, (3) the duration of a time slot, and
- (4) the duration of a frame.
- **Solution**
- **1.** The duration of 1 bit before multiplexing is 1/1 kbps, or 0.001 s (1 ms).
- **2.** The rate of the link is 4 times the rate of a connection, or 4 kbps.
- **3.** The duration of each time slot is one-fourth of the duration of each bit before multiplexing,
- or 1/4 ms or 250 µs. Note that we can also calculate this from the data rate of the link, 4 kbps.
- The bit duration is the inverse of the data rate, or 1/4 kbps or 250 µs.
- **4.** The duration of a frame is always the same as the duration of a unit before multiplexing, or
- 1 ms. We can also calculate this in another way. Each frame in this case has four time slots.
- So the duration of a frame is 4 times 250 µs, or 1 ms.

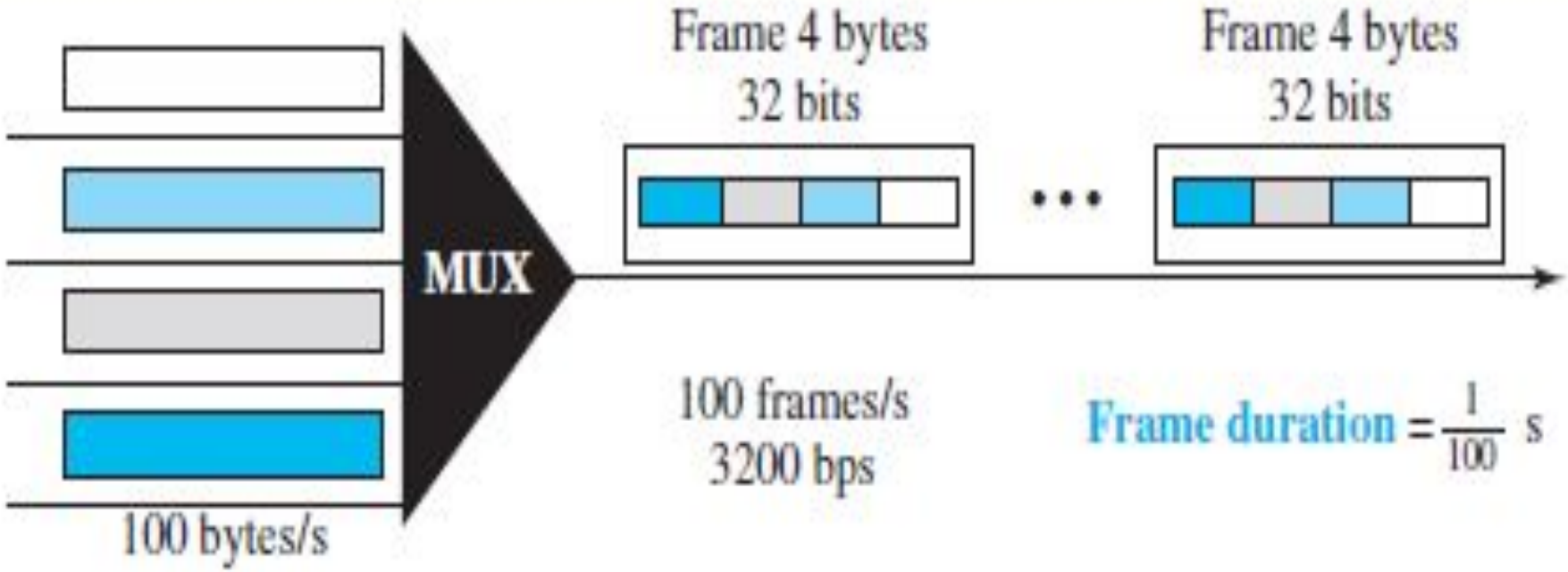
- ***Interleaving***
- TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side.
- The switches are synchronized and rotate at the same speed, but in opposite directions.
- On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path.
- This process is called **interleaving.**
- On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.

Figure 6.15 *Interleaving*



- Example 6.8
- Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link
- **Solution**
- The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits.
- Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second.
- The duration of a frame is therefore 1/100 s. The link is carrying 100 frames per second, and since each frame contains 32 bits, the bit rate is 100 × 32, or 3200 bps. This is actually 4 times the bit rate of each channel, which is 100 × 8 = 800 bps.

Figure 6.16 *Example 6.8*



- **Example 6.9**
- A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate?What is the bit duration?
- **Solution**
- Figure 6.17 shows the output for four arbitrary inputs. The link carries 50,000 frames per second since each frame contains 2 bits per channel. The frame duration is therefore 1/50,000 s or 20 µs.
- The frame rate is 50,000 frames per second, and each frame carries 8 bits; the bit rate is 50,000 × 8 = 400,000 bits or 400 kbps. The bit duration is 1/400,000 s, or 2.5 µs.
- frame duration is 8 times the bit duration because each frame is carrying 8 bits.

## *Empty Slots*

- Synchronous TDM is not as efficient as it could be. If a source does not have data to send, the corresponding slot in the output frame is empty..
- The first output frame has three slots filled, the second frame has two slots filled, and the third frame has three slots filled. No frame is full. statistical TDM can improve the efficiency by removing the empty slots from the frame.

Figure 6.17 *Example 6.9*

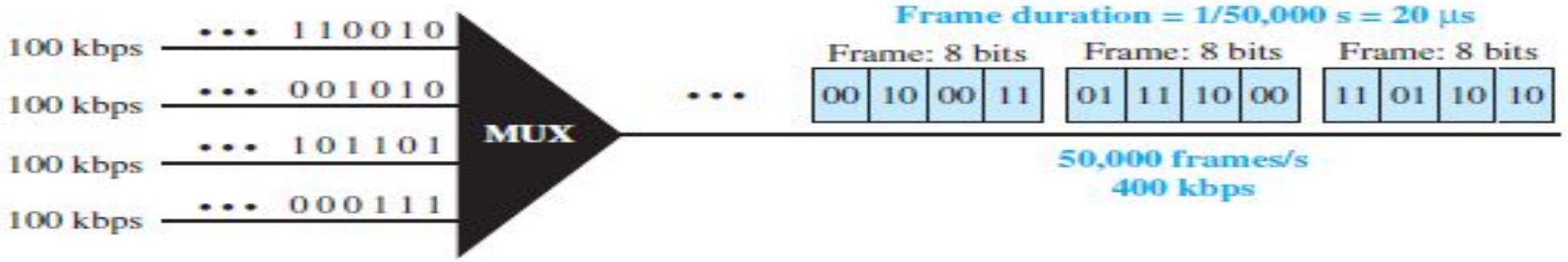
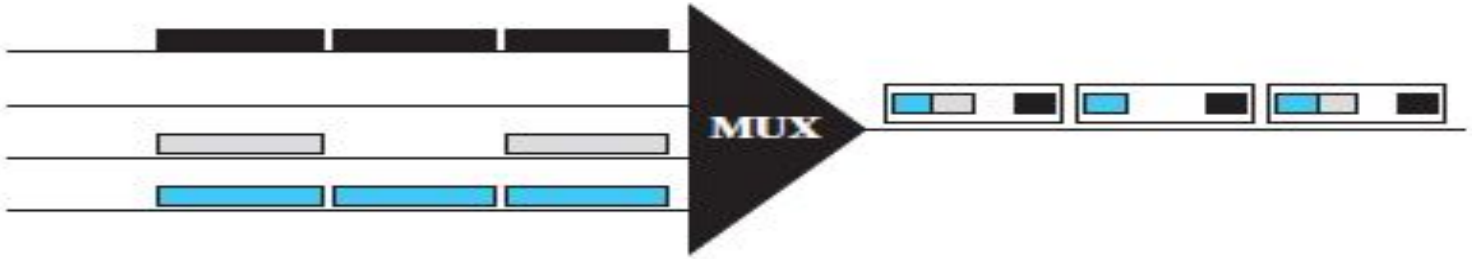
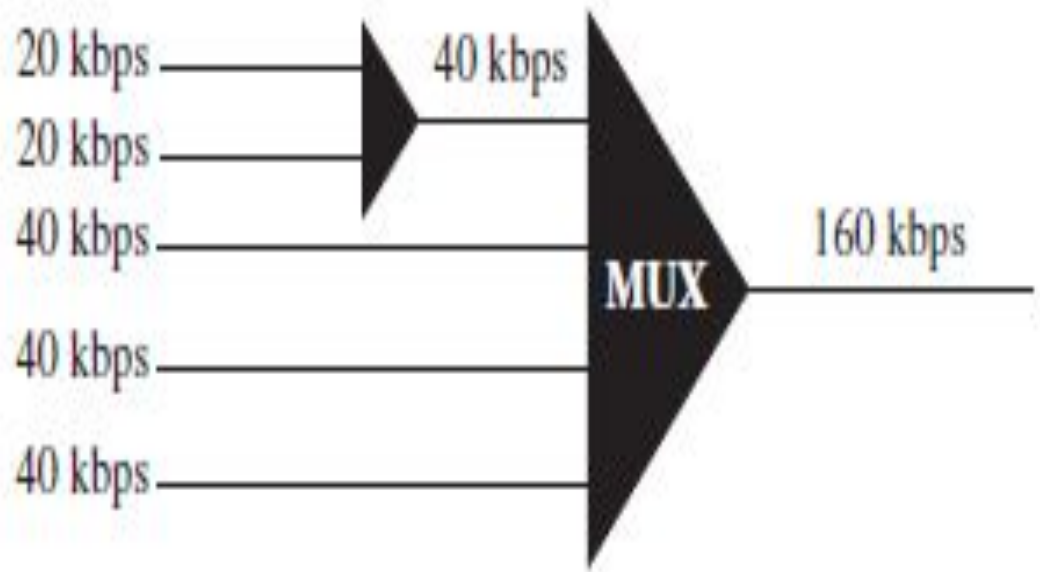



Figure 6.18 *Empty slots*



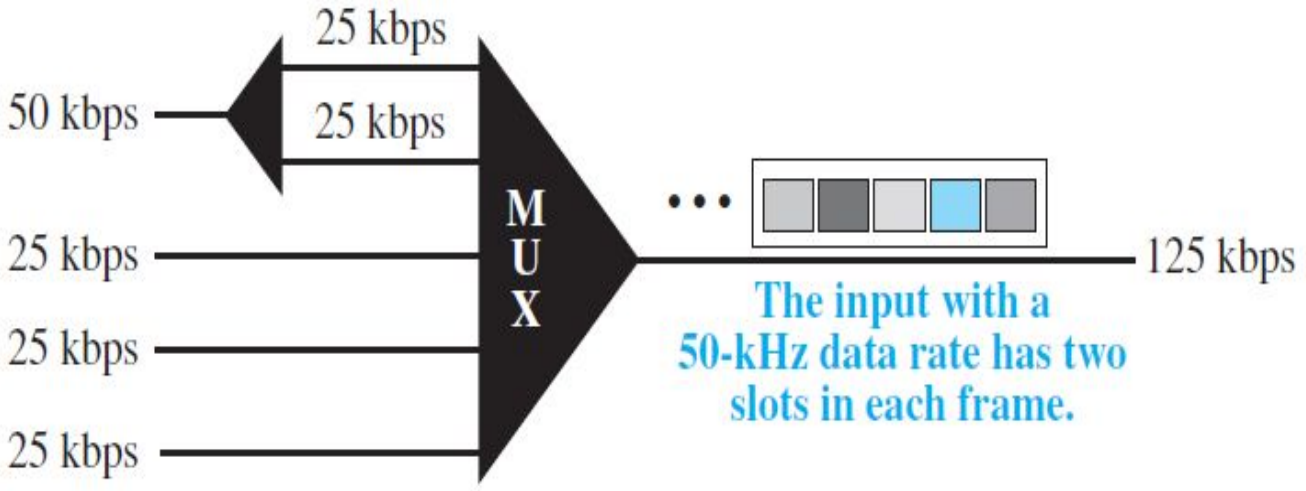
- ***Data Rate Management***
- One problem with TDM is how to handle a disparity in the input data rates.
- if data rates are not the same, three strategies, or a combination of them, can be used.
- We call these three strategies **multilevel multiplexing, multiple-slot allocation,** and **pulse stuffing.**
- ***Multilevel Multiplexing*** Multilevel multiplexing is a technique used when the data rate of an input line is a multiple of others
- Ex: we have two inputs of 20 kbps and three inputs of 40 kbps. The first two input lines can be multiplexed together to provide a data rate equal to the last three. A second level of multiplexing can create an output of 160 kbps.

Figure 6.19 *Multilevel multiplexing*



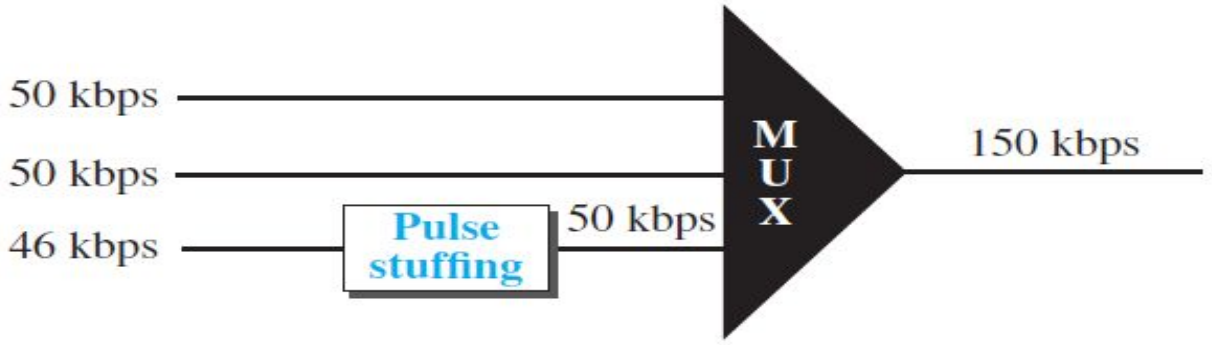
- ***Multiple-Slot Allocation*** Sometimes it is more efficient to allot more than one slot in a frame to a single input line.
- Ex:  we might have an input line that has a data rate that is a multiple of another input. In Figure the input line with a 50-kbps data rate can be given two slots in the output. We insert a demultiplexer in the line to make two inputs out of one.

**Figure 6.20** *Multiple-slot multiplexing*



- ***Pulse Stuffing*** Sometimes the bit rates of sources are not multiple integers of each other.
- Therefore, neither of the above two techniques can be applied. One solution is to make the highest input data rate the dominant data rate and then add dummy bits to the input lines with lower rates. This will increase their rates. This technique is called ***pulse stuffing,*** *bit padding,* or *bit stuffing.*
- The input with a data rate of 46 is pulse-stuffed to increase the rate to 50 kbps. Now multiplexing can take place.

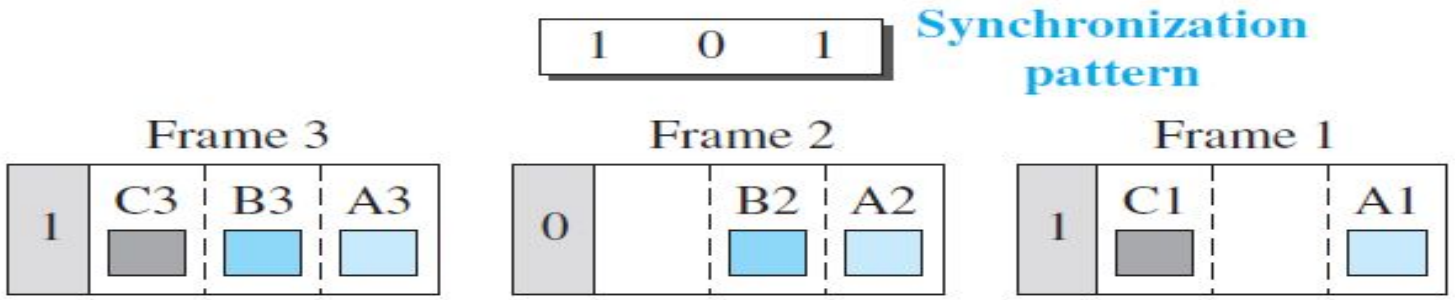
**Figure 6.21** *Pulse stuffing*



## *Frame Synchronizing*

- The implementation of TDM is not as simple as that of FDM. Synchronization between
- If the multiplexer and the demultiplexerare not synchronized, a bit belonging to one channel may be received by the wrong channel. For this reason, one or more synchronization bits are usually added to the beginning of each frame. These bits, called **framing bits,** follow a pattern frame to frame, that allows the demultiplexer to synchronize with the incoming stream so that
- it can separate the time slots accurately. In most cases, this synchronization information consists of 1 bit per frame, alternating between 0 and 1

**Figure 6.22** *Framing bits*



- **Example 6.10**
- We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (1) the data rate of each source, (2) the duration of each character in each source, (3) the frame rate, (4) the duration of each frame, (5) the number of bits in each frame, and (6) the data rate of the link.
- **Solution**:

**1.** The data rate of each source is 250 × 8 = 2000 bps = 2 kbps.

**2.** Each source sends 250 characters per second; therefore, the duration of a character is 1/250 s, or 4 ms.

**3.** Each frame has one character from each source, which means the link needs to send 250 frames per second to keep the transmission rate of each source.

**4.** The duration of each frame is 1/250 s, or 4 ms. Note that the duration of each frame is the same as the duration of each character coming from each source.

**5.** Each frame carries 4 characters and 1 extra synchronizing bit. This means that each frame is 4 × 8 + 1 = 33 bits.

**6.** The link sends 250 frames per second, and each frame contains 33 bits. This means that the data rate of the link is 250 × 33, or 8250 bps.

Note that the bit rate of the link is greater than

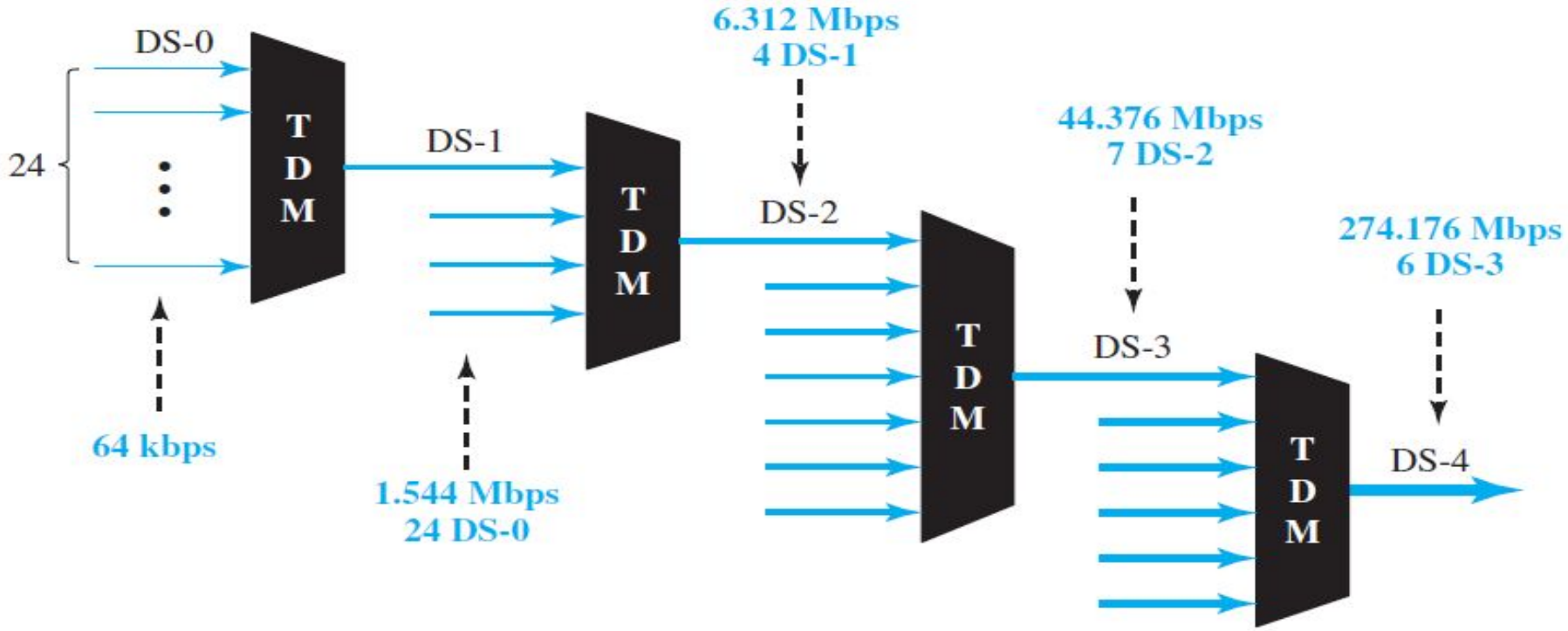
- the combined bit rates of the four channels. If we add the bit rates of four channels, we get
- 8000 bps. Because 250 frames are traveling per second and each contains 1 extra bit for synchronizing, we need to add 250 to the sum to get 8250 bps.

- Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?
- **Solution**
- We can allocate one slot to the first channel and two slots to the second channel. Each frame carries 3 bits. The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. The frame duration is 1/100,000 s, or 10 ms.
- The bit rate is 100,000 frames/s × 3 bits per frame, or 300 kbps. Note that because each frame carries 1 bit from the first channel, the bit rate for the first channel is preserved.
- The bit rate for the second channel is also preserved because each frame carries 2 bits from the second channel.

# *Digital Signal Service*

- Telephone companies implement TDM through a hierarchy of digital signals, called ***digital signal (DS) service*** or *digital hierarchy*.

**Figure 6.23** *Digital hierarchy*



- **DS-0** is a single digital channel of 64 kbps.

❑ **DS-1** is a 1.544-Mbps service; 1.544 Mbps is 24 times 64 kbps plus 8 kbps of overhead.

- It can be used as a single service for 1.544-Mbps transmissions, or it can be
- used to multiplex 24 DS-0 channels or to carry any other combination desired by
- the user that can fit within its 1.544-Mbps capacity.

❑ **DS-2** is a 6.312-Mbps service; 6.312 Mbps is 96 times 64 kbps plus 168 kbps of overhead. It can be used as a single service for 6.312-Mbps transmissions; or it can be used to multiplex 4 DS-1 channels, 96 DS-0 channels, or a combination of these service types.

❑ **DS-3** is a 44.376-Mbps service; 44.376 Mbps is 672 times 64 kbps plus 1.368 Mbps

- of overhead. It can be used as a single service for 44.376-Mbps transmissions; or it
- can be used to multiplex 7 DS-2 channels, 28 DS-1 channels, 672 DS-0 channels,
- or a combination of these service types.

❑ **DS-4** is a 274.176-Mbps service; 274.176 is 4032 times 64 kbps plus 16.128 Mbps of

- overhead. It can be used to multiplex 6 DS-3 channels, 42 DS-2 channels, 168 DS-1
- channels, 4032 DS-0 channels, or a combination of these service types.

# *T Lines*

- DS-0, DS-1, and so on are the names of services. To implement those services, the telephone companies use **T lines** (T-1 to T-4). These are lines with capacities precisely matched to the data rates of the DS-1 to DS-4 services So far only T-1 and T-3 lines are commercially available.

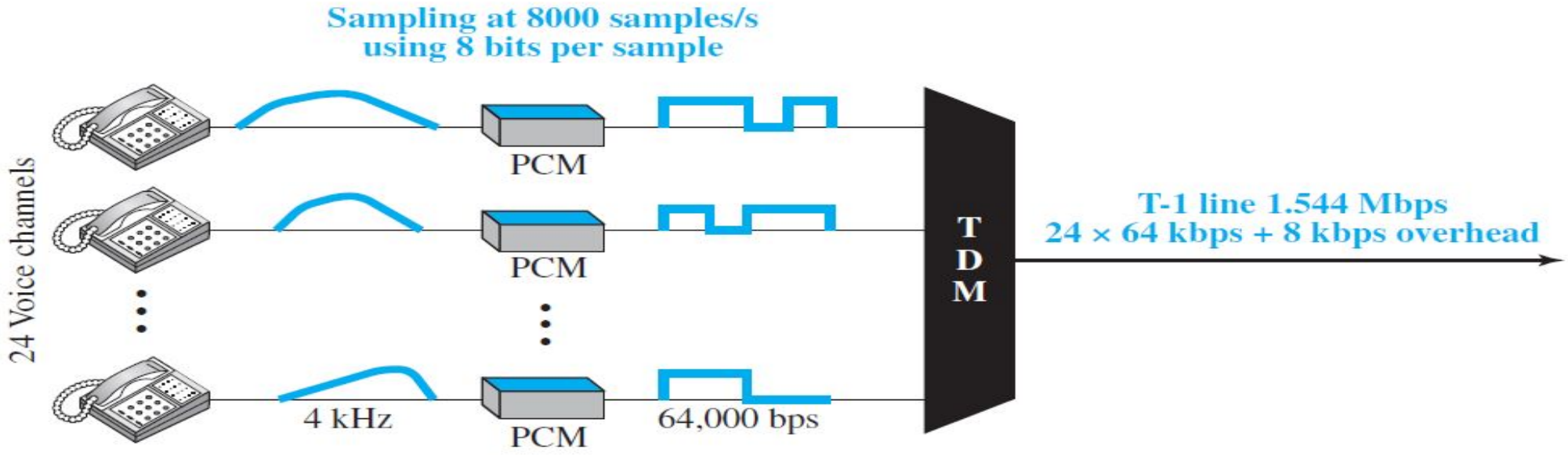
**Table 6.1** *DS and T line rates*

| *Service* | *Line* | *Rate (Mbps)* | *Voice Channels* |
|---|---|---|---|
| DS-1 | T-1 | 1.544 | 24 |
| DS-2 | T-2 | 6.312 | 96 |
| DS-3 | T-3 | 44.736 | 672 |
| DS-4 | T-4 | 274.176 | 4032 |

# *T Lines for Analog Transmission*

- T lines are digital lines designed for the transmission of digital data, audio, or
- video. they also can be used for analog transmission (regular telephone
- connections), provided the analog signals are first sampled, then time-division multiplexed.

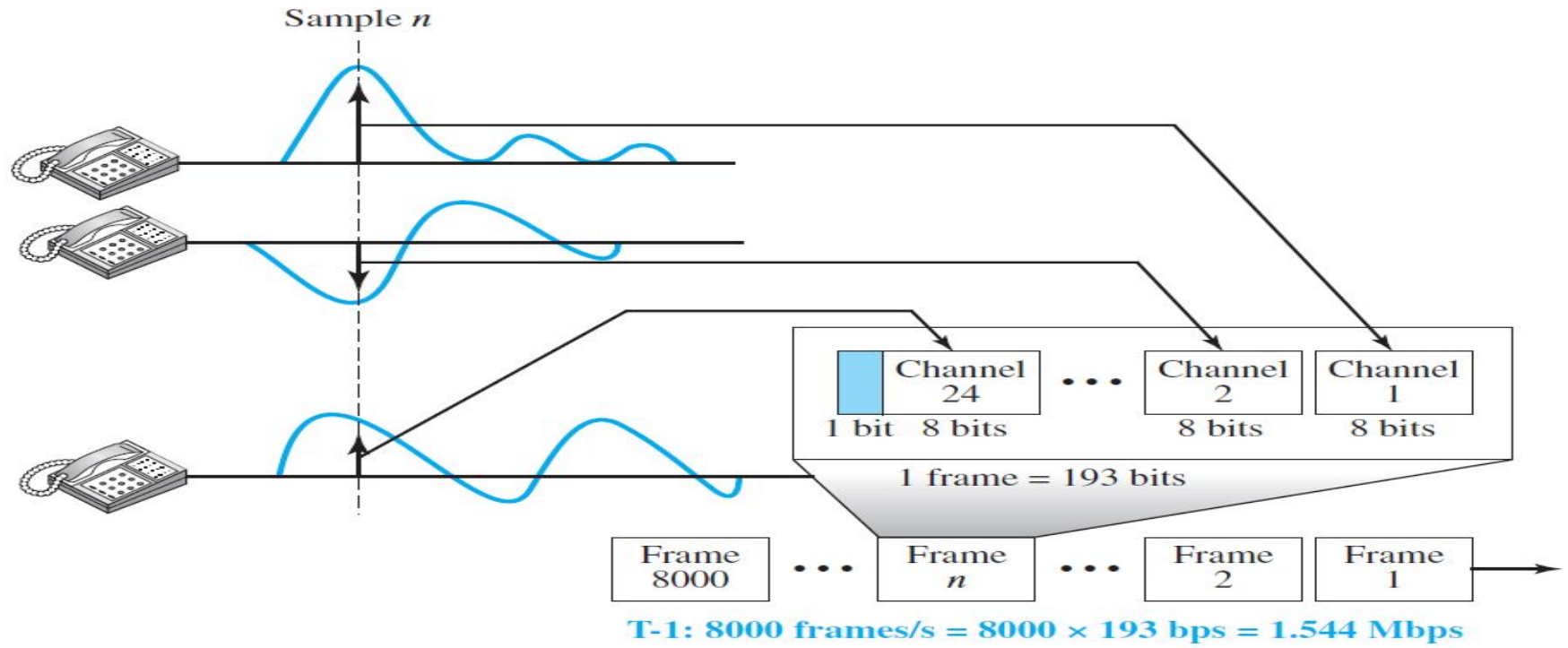
**Figure 6.24** *T-1 line for multiplexing telephone lines*



# *The T-1 Frame:*

- ***The T-1 Frame:*** DS-1 requires 8 kbps of overhead. The frame used on a T-1 line is usually 193 bits divided into 24 slots of 8 bits each plus 1 extra bit for synchronization (24 × 8 + 1 = 193); each slot contains one signal segment from each channel; 24 segments are interleaved in one frame. If a T-1 line carries 8000 frames, the data rate is 1.544 Mbps (193 × 8000 = 1.544 Mbps)—the capacity of the line.

**Figure 6.25** *T-1 frame structure*



- ***E Lines***
- Europeans use a version of T lines called **E lines.** The two systems are conceptually identical, but their capacities differ

**Table 6.2** *E line rates*

| *Line* | *Rate (Mbps)* | *Voice Channels* |
|---|---|---|
| E-1 | 2.048 | 30 |
| E-2 | 8.448 | 120 |
| E-3 | 34.368 | 480 |
| E-4 | 139.264 | 1920 |

# *More Synchronous TDM Applications*

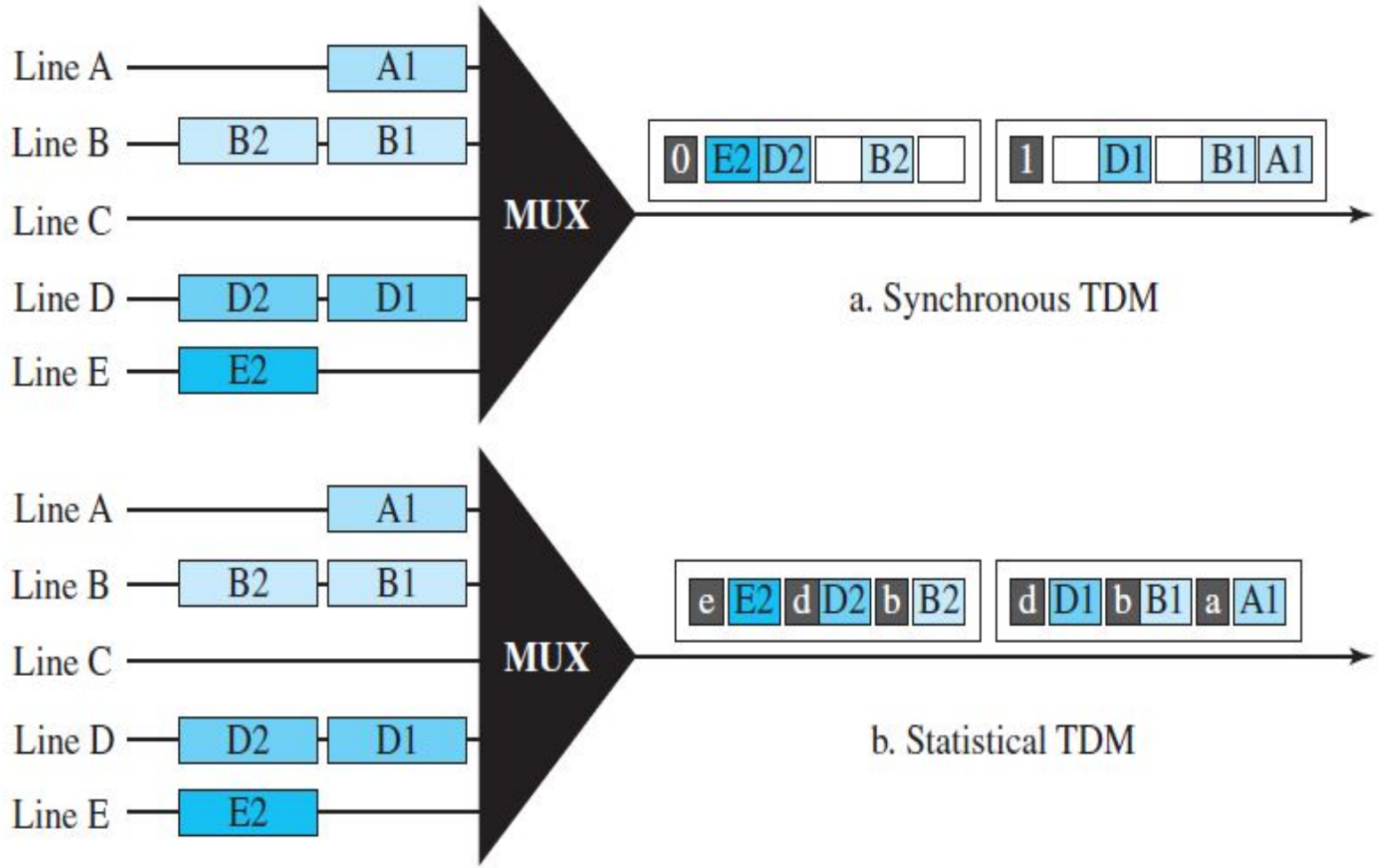
- Some second-generation cellular telephone companies use synchronous TDM.
- Example: the digital version of cellular telephony divides the available bandwidth into30-kHz bands.
- For each band, TDM is applied so that six users can share the band.

- This means that each 30-kHz band is now made of six time slots, and the digitized voice signals of the users are inserted in the slots.

- Using TDM, the number of telephone users in each area is now 6 times greater

- ***Statistical Time-Division Multiplexing***
- in synchronous TDM, each input has a reserved slotMin the output frame. This can be inefficient if some input lines have no data to send
- statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.
- Only when an input line has a slot's worth of data to send is it given a slot in the output frame.
- In statistical multiplexing, the number of slots in each frame is less than the number of input lines.
- The multiplexer checks each input line in roundrobin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.

**Figure 6.26** *TDM slot comparison*



- ***Addressing***
- An output slot in synchronous TDM is totally occupied by data; in statistical TDM, a slot needs to carry data as well as the address of the destination.
- In synchronous TDM, there is no need for addressing; synchronization and preassigned relationships between the inputs and outputs serve as an address.
- EX: that input 1 always goes to input 2. If the multiplexer and the demultiplexer are synchronized, this is guaranteed.
- In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots.
- We need to include the address of the receiver inside each slot to show where it is to be delivered. The addressing in its simplest form can be *n* bits to define *N* different output lines with $n = \log_2 N$. For example, for eight different output lines, we need a 3-bit address.

- ***Slot Size***
- Since a slot carries both data and an address in statistical TDM, the ratio of the data size to address size must be reasonable to make transmission efficient.
- Ex: it would be inefficient to send 1 bit per slot as data when the address is 3 bits. This would mean an overhead of 300 percent. In statistical TDM, a block of data is usually many bytes while the address is just a few bytes.
- ***No Synchronization Bit***
- There is another difference between synchronous and statistical TDM, but this time it is at the frame level. The frames in statistical TDM need not be synchronized, so we do not need synchronization bits.

- ***Bandwidth***

- In statistical TDM, the capacity of the link is normally less than the sum of the capacities of each channel.
- The designers of statistical TDM define the capacity of the link based on the statistics of the load for each channel.
- If on average only *x* percent of the input slots are filled, the capacity of the link reflects this. Of course, during peak times,some slots need to wait.

# SPREAD SPECTRUM

- Multiplexing combines signals from several sources to achieve bandwidth efficiency;

the available bandwidth of a link is divided between the sources.

- In **spread spectrum (SS),** we also combine signals from different sources to fit into a larger bandwidth.
- Spread spectrum is designed to be used in wireless applications (LANs and WANs).
- In wireless applications, all stations use air (or a vacuum) as the medium for communication.
- Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder

- **1.** The bandwidth allocated to each station needs to be, by far, larger than what is needed. This allows redundancy.
- The expanding of the original bandwidth *B* to the bandwidth *B*ss must be done by a process that is independent of the original signal. In other words, the spreading process occurs after the signal is created by the source. After the signal is created by the source, the spreading process uses a spreading code and spreads the bandwidth.
- The spreading code is a series of numbers that look random, but are actually a pattern.

**Figure 6.27** *Spread spectrum*

